

Palestine Violence Archive

Dashboard & Pipeline User Guide

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This guide explains how to use the Streamlit pipeline dashboard, what each page does, how research assistants should review batches, and how to export data for statistical analysis.

1. Project Overview

The Palestine Violence Archive is a research pipeline built for a political science lab at UC San Diego. It automatically collects news articles about human rights incidents affecting Palestinians in the West Bank from the WAFA news agency, classifies each incident across 10 violence labels using machine learning, routes uncertain cases to human reviewers, and continuously retrains the model as reviewed labels accumulate.

The goal is to build a clean, labeled dataset of incidents (raids, arrests, property destruction, etc.) that the professor and research assistants (RAs) can export to SPSS or R for statistical analysis and eventual publication.

2. Pipeline Flow

The pipeline runs in four stages:

1. Ingest - Scrapes new WAFA articles and writes them to the Google Sheets 'Reports' tab. Duplicate URLs are skipped automatically.
2. Export - Reads unprocessed rows from the 'Reports' tab and saves them as a timestamped CSV in data/incoming/.
3. Predict - Runs the baseline (TF-IDF + logistic regression) or DeBERTa v3 model on the batch CSV. Produces a compact review sheet (XLSX/CSV) in data/review_exports/ with model predictions, confidence scores, uncertainty flags, and empty human_* columns for RAs to fill.
4. Retrain - After RAs review enough batches and labels are merged back into the training dataset, the baseline model is rebuilt and evaluated. DeBERTa can then be fine-tuned separately.

3. Dashboard Pages

The left sidebar lists five pages. Each is described below.

3a. Dashboard

The landing page gives a live snapshot of the pipeline state.

Incoming batch size - Number of articles waiting in data/incoming/ that have not yet been reviewed.

Review queue - How many rows in the latest review sheet still have review_status != 'reviewed'.

Model metrics - Per-label F1, precision, and recall from the last baseline training run (read from baseline_meta.json).

IRR panel - Computes inter-rater reliability (Cohen's Kappa) for double-reviewed rows. A macro Kappa above 0.80 indicates strong agreement between RAs. Click 'Compute IRR' to refresh after RAs finish a batch.

Refresh - Clicking 'Refresh' reloads all metrics without re-running the pipeline.

3b. Run Pipeline

The main control panel for running the full ingest -> export -> predict workflow.

Model - Choose 'baseline' (fast, CPU-only, good for daily runs) or 'deberta' (slower, GPU-recommended, higher accuracy).

Skip ingest - Check this if you do not want to scrape new articles. The pipeline will use whatever is already in Google Sheets.

Re-export all reports - Forces re-export of all reports, ignoring the already-exported tracking file. Use if you need to reprocess old data.

Push compact review sheet to Sheets - After predicting, automatically uploads the review sheet to the Google Sheets 'Review' tab so RAs can work collaboratively online.

Dry run - Prints exactly what each step would do without writing any files. Always a safe first step on a new machine.

Deduplicate batch - Removes near-duplicate articles (same date and governorate, very similar text) before prediction, preventing double-counting.

Max rows per batch - Caps the batch at N rows. Recommended: 100-150 to keep review sessions manageable for RAs.

Quick Actions - Pre-configured shortcuts for the most common runs (skip ingest + baseline + push to Sheets; dry run preview).

Log output - Streams live as the pipeline runs. The log persists on screen after the run completes so you can scroll through it.

3c. Review Access

Provides access to the latest batch files and links to Google Sheets.

Google Sheets link - Opens the shared spreadsheet where RAs review predictions. The 'Reports' tab contains raw scraped articles; the 'Review' tab contains model predictions with empty `human_*` columns.

Latest review file - Shows the most recent `review_compact_*.xlsx` or `.csv` from `data/review_exports/`. Click to preview the first 20 rows.

Latest predictions file - Shows the raw inference audit trail (all confidence scores and flags) from the most recent predict run.

RA workflow reminder - Step-by-step instructions reminding RAs what to fill in (`human_*` columns, `review_status = 'reviewed'`, `reviewer_notes`) and where to save the completed file.

3d. Export Analysis

Exports the cleaned, labeled incident database to formats the professor can open directly in SPSS, JASP, or R.

Years - Which years of processed data to include (default: 2023 2024 2025). Requires `data/processed/oppression_{year}.csv` to exist; run `normalize_raw.py` first if those files are missing.

Output directory - Where to write the exported files (default: `analysis/`).

Skip SPSS export - Skips writing the `.sav` file. Use this if `pyreadstat` is not installed.

Exported files - After running, the page lists all exported files with sizes. The `.sav` file includes full variable labels and value labels (e.g. 0 = No, 1 = Yes for all binary columns). The `codebook_variables.csv` is a

variable dictionary showing type, valid N, missing %, and value ranges.

R usage - An R code snippet at the bottom shows how to load the .sav with value labels (haven::read_sav) or the plain CSV, and how to run the bundled descriptive_analysis.R script that produces 11 summary tables and 7 publication-ready figures.

3e. Retrain (Admin only)

This page rebuilds the active model. Only run it after enough reviewed batches have been merged into the training dataset.

Training data stats - Shows the current row count of master_reviewed_dataset.csv so you know how much labeled data exists.

Checkpoint system - Before retraining, the dashboard automatically creates a timestamped backup of all model artifacts (baseline .pkl files, DeBERTa meta JSON, master CSV). You can also create a manual checkpoint at any time and restore any previous checkpoint if something goes wrong.

Retrain baseline - Runs export_training_data.py followed by train_baseline_multilabel.py. Takes 1-5 minutes on CPU.

Retrain DeBERTa - Fine-tunes the DeBERTa v3 transformer. Requires a GPU; plan for 1-3 hours on a single card.

Merge reviewed labels - Merges a completed review XLSX back into the master training CSV. Run this after RAs finish each batch before retraining. Automatically checkpoints before merging.

4. RA Review Workflow

Research assistants follow this process for each batch:

Step 1 - Open the review sheet. Either use the Google Sheets link (Review tab) or download the .xlsx from data/review_exports/.

Step 2 - For each row, read the description column and check or correct the 10 pred_* columns. Fill in the corresponding human_* column with 1 (yes) or 0 (no). Leave blank if uncertain and add a note.

Step 3 - For rows marked double_review = TRUE, a second RA should independently fill the human2_* columns without looking at the first RA's answers. This enables inter-rater reliability measurement.

Step 4 - Set review_status = 'reviewed' for each completed row. Optionally fill reviewer_notes with observations.

Step 5 - Save the completed file to data/reviewed/ and notify the pipeline operator to run the merge step in the Retrain page.

5. Incident Labels

The model classifies each incident into 10 binary labels (1 = present):

Raid - Organized entry by Israeli forces into a Palestinian area to arrest, search, or impose presence.

Arrest/Detention - Apprehension or detention of Palestinians by Israeli forces or the Palestinian Authority.

Physical Assault - Direct violence: shooting, beating, tear gas, stun grenades, dogs.

Coercive Actions - Intimidation or harassment without direct assault: verbal threats, searches, show-of-force deployments.

Restriction of Freedoms - Curfews, checkpoints, roadblocks, prevention of agricultural or medical access.

Religious Encroachment - Incursion into or desecration of mosques, churches, or holy sites; interference with prayers or observances.

Harm to Property - Vandalism (graffiti, slashed tires) or destruction (demolition, arson) of Palestinian property.

Dispossession - Seizure or theft of land, livestock, crops, or assets.

Protest - Palestinian organized demonstration, march, or civil resistance action.

Multi-Community Incident - Simultaneous or coordinated action targeting multiple Palestinian villages or towns.

6. Models

Baseline (TF-IDF + Logistic Regression)

Location: modeling/saved_models/baseline_rank_based_fixed/

Speed: seconds on CPU. Best for daily runs and quick iteration.

Thresholding: rank-based - the model predicts the top-k articles per label, where k matches the observed training prevalence rate.

DeBERTa v3 (Transformer)

Location: modeling/saved_models/deberta_v3_multilabel/

Weights: stored on HuggingFace Hub (jalva182/palestine-violence-deberta)

Speed: ~1 min per 100 rows on GPU; much slower on CPU.

Use for final-pass labeling when accuracy matters most.

Both models output a confidence score (0-1) per label. Rows where confidence is within 0.10 of the decision threshold are flagged as needs_review = TRUE and routed to human reviewers.

7. File Locations

data/incoming/ - Raw batches exported from Google Sheets
data/review_exports/ - Model predictions + compact review sheets
data/reviewed/ - Completed RA review files
data/training/ - Master labeled dataset (used for retraining)
data/processed/ - Normalized yearly CSVs (2023, 2024, 2025)
analysis/ - SPSS .sav, R CSV, codebook, R analysis script
modeling/saved_models/ - Baseline and DeBERTa model artifacts
modeling/backups/ - Timestamped model checkpoints
logs/ - Pipeline run logs (pipeline_YYYYMMDD.log)
config/ - Label codebook YAML

8. Common Issues

No articles scraped (0 new reports)

Cause: robots.txt block or network issue.

Fix: Check the log for 'robots.txt' warnings. Try --skip-ingest and re-export from what is already in Sheets.

pyreadstat not installed

Fix: pip install pyreadstat (or check 'Skip SPSS export' on the Export Analysis page).

DeBERTa weights not found

Fix: Download from HuggingFace Hub into
modeling/saved_models/deberta_v3_multilabel/deberta_v3_best/

Google Sheets authentication error

Fix: Ensure the service account JSON key path is set in environment variable
GOOGLE_APPLICATION_CREDENTIALS, or in Streamlit secrets.

Log disappears when clicking buttons

This is fixed - logs are stored in session state and persist across page interactions.

